Detailed HIL test procedures

STM32 Nucleo-G431KB / Zephyr / DRV8871 | 26 procedures | Fillable checkboxes

Run ID / date / operator

Firmware commit / binary hash / board serial

Motor / encoder / load / fixture record

Approved requirements / limits revision

Using this document

Each checkbox records an action completed, not a passing outcome. Select the procedure result separately. Use a separate copy or evidence record for each configuration and repetition.

H = real-controller HIL with a closed-loop real-time plant model. P = powered motor/load bench. H + P requires both. Missing implementation or an unapproved required limit is Blocked/TBD, never Pass.

Read the approved protocol and HIL requirements before testing. Repeat every procedure over its required conditions and retain failed/aborted results. Do not treat a simulator-only result as proof of physical current, thermal or stopping performance.

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Readiness and acceptance limits

Complete the shared prerequisites before executing any procedure. Unresolved values stay TBD.

- [ ] Approve the command protocol, transition matrix and fresh-command/restart policy.
- [ ] Identify safe output levels for each startup, stop and fault case; zero duty alone is insufficient.
- [ ] Verify timestamp alignment, calibration and uncertainty; qualify the HIL model and check for overruns.
- [ ] Verify independent fixture shutdown, current/thermal/energy limits and guarding before powered cases.
- [ ] Record production logging, instrumentation overhead and any test hooks; establish a baseline production-build run.

E - Supply/current/regen envelope; exposure and abort thresholds

M - Speed/load/inertia; stop speed, time and travel

C - Rise/settle time, error, overshoot, ripple and recovery

T - Boot/command/STOP/schedule deadlines; loop jitter

F - Command-loss, feedback, stall and watchdog deadlines

S - Coast/brake/disable; cancellation and restart policy

B - CAN rates/bursts; response and overflow behavior

Q - Operating corners, samples, repetitions and duration

U - Measurement/model uncertainty and decision guard bands

Enter values or a controlled limits-record reference. All relevant limits must be fixed before acceptance. For an upper bound, use measured maximum plus uncertainty <= limit, or another approved decision rule.

HIL-001 | Startup and reset behavior

Method: H + P | Repeat across the approved Q matrix; use the signed-off E/M/C/T/F/S/B/U limits.

SETUP / WHAT TO WATCH

Monitor supply/reset, PB6/PB7 and driver inputs, motor current, encoder/shaft motion, CAN traffic and readiness indication; capture before power is applied.

ACTIONS TO COMPLETE

- [ ] 1. Approve startup input levels, boot deadline, allowed movement/current and startup-command acceptance policy; record supply sequences and repetitions.
- [ ] 2. Power up with the CAN host disconnected. Verify the approved inactive output and absence of unintended motion.
- [ ] 3. Repeat with the host connected but silent. Verify bus presence does not authorize movement.
- [ ] 4. Transmit commands before and during initialization. Verify the approved acceptance/rejection policy and that stale commands do not execute later.
- [ ] 5. Power the MCU before the driver supply, then reverse the sequence. Capture both transitions and verify startup limits throughout.
- [ ] 6. Reset the MCU while idle, then during controlled low-energy motion. Verify the defined reset stop response and absence of automatic restart.
- [ ] 7. Repeat the approved power-cycle and off-time matrix. Measure worst-case readiness time, current and shaft displacement.
- [ ] 8. After readiness, issue a fresh valid motion command, followed by STOP. Confirm normal authorized operation and attach the synchronized traces.

PASS CRITERIA

Measured inputs and motor output remain within S/M limits until initialization is complete and fresh authorization is accepted; ready state reached within T. Scope capture includes the reset interval before firmware runs.

Result Not Run

Run / method / conditions

Evidence / measurements / deviations / N/A justification

Operator / date

Reviewer / date

HIL-002 | Direction, speed sign and zero command

Method: H + P | Repeat across the approved Q matrix; use the signed-off E/M/C/T/F/S/B/U limits.

SETUP / WHAT TO WATCH

Capture decoded commands, PWM/driver outputs, independent shaft direction/speed, encoder sign and current. Define the positive shaft direction physically.

ACTIONS TO COMPLETE

- [ ] 1. Approve speed points, direction/sign encoding, zero-command behavior and direction/current limits before execution.
- [ ] 2. From the defined ready state, command minimum nonzero forward speed. Confirm physical direction, encoder sign and speed agree with the specification.
- [ ] 3. Repeat at middle and maximum supported forward speeds across the approved load/supply cases.
- [ ] 4. Return to the defined stop state; repeat the same points in reverse. Confirm direction and feedback sign again.
- [ ] 5. Apply the supported sign/direction combinations. Verify consistent interpretation; reject unsupported combinations according to the protocol.
- [ ] 6. Command zero from both directions. Verify the approved coast/brake/stop behavior and permissible residual motion.
- [ ] 7. Repeat near the minimum controllable speed to expose quantization or direction chatter. Compare errors with approved near-zero absolute limits.
- [ ] 8. Record direction agreement, maximum speed error and zero-command response with trace references for every operating point.

PASS CRITERIA

Requested direction, measured shaft direction, and feedback sign agree; zero command implements the defined S behavior; speed stays within M/C limits.

Result Not Run | Run / method / conditions

Evidence / measurements / deviations / N/A justification

Operator / date | Reviewer / date

HIL-003 | PWM waveform and drive modes

Method: H + P | Repeat across the approved Q matrix; use the signed-off E/M/C/T/F/S/B/U limits.

SETUP / WHAT TO WATCH

Use synchronized logic and differential voltage capture at MCU/driver inputs and motor output. Observe actual input polarity, current and supply voltage.

ACTIONS TO COMPLETE

- [ ] 1. Approve frequency, duty range, minimum pulse width, driver truth table, drive modes and transient limits for the exact board.
- [ ] 2. Begin with HIL or a low-energy fixture. Verify commanded inactive, forward, reverse, brake and coast states against actual input/output levels.
- [ ] 3. Sweep output commands from minimum nonzero to near maximum in both directions; include zero and supported endpoints.
- [ ] 4. Measure PWM frequency, effective duty and shortest high/low pulses. Account for polarity inversion and driver input timing requirements.
- [ ] 5. Capture mode transitions and direction changes. Check for unintended pulses, unexpected drive polarity and excessive current.
- [ ] 6. Repeat at approved supply/load conditions and production logging settings. Confirm the control-to-PWM mapping remains bounded.
- [ ] 7. Where direct duty injection needs a test hook, document it; repeat representative cases through normal CAN commands using production firmware.
- [ ] 8. Save waveform captures and worst-case frequency, duty, pulse-width and transient measurements; compare each with approved limits.

PASS CRITERIA

Frequency, duty, pulse widths, and driver output match the approved electrical specification; no unintended direction pulse. Check the DRV8871 input timing and truth table, not just MCU duty percentages. TI datasheet, sections 6.3 and 7.3.1.

Result Not Run

Run / method / conditions

Evidence / measurements / deviations / N/A justification

Operator / date

Reviewer / date

HIL-004 | PI/PID speed step response

Method: H + P | Repeat across the approved Q matrix; use the signed-off E/M/C/T/F/S/B/U limits.

SETUP / WHAT TO WATCH

Capture setpoint, independent shaft speed, encoder-derived speed, PWM/output, current, bus voltage and timestamps. Record gains, filters and measured loop timing.

ACTIONS TO COMPLETE

- [ ] 1. Freeze PI/PID mode, gains, limits, rise-time definition, settling band/window, overshoot limit and steady-error window. Separate tuning from acceptance runs.
- [ ] 2. Establish a stable low-speed operating point. Apply a specified upward speed step without changing gains; record before and after the step.
- [ ] 3. Repeat downward steps and steps between approved low, middle and high speeds. Repeat both directions and the approved supply/load matrix.
- [ ] 4. Measure 10-90% rise time for applicable steps, peak overshoot, settling time and steady-state error using the approved definitions.
- [ ] 5. Check near-zero speed separately using absolute error and ripple limits; do not normalize error by a zero reference.
- [ ] 6. Evaluate oscillation, output saturation and current/voltage peaks throughout each response. All must remain inside the approved envelopes.
- [ ] 7. Repeat representative cases at maximum supported CAN/CPU load. Record measured sample period and jitter with each response.
- [ ] 8. Complete the controller response worksheet, attach raw traces and retain worst-case results. For derivative-enabled builds, also repeat noise-response cases.

PASS CRITERIA

Steady error, settling time, overshoot, current, and output stay within C/E/M. Independent shaft-speed measurement corroborates P results.

Result Not Run

Run / method / conditions

Evidence / measurements / deviations / N/A justification

Operator / date

Reviewer / date

HIL-005 | Load disturbance and anti-windup recovery

Method: H + P | Repeat across the approved Q matrix; use the signed-off E/M/C/T/F/S/B/U limits.

SETUP / WHAT TO WATCH

Use a closed-loop HIL plant, then a controlled motor/load fixture. Record target/speed, torque stimulus, output, current, voltage and integral state when observable.

ACTIONS TO COMPLETE

- [ ] 1. Approve load-step size, saturation exposure time, recovery/overshoot limits and abort thresholds; hold gains fixed for acceptance.
- [ ] 2. At constant speed, apply a defined load increase. Measure peak speed deviation and time to recover into the approved band.
- [ ] 3. Remove that load and repeat. Check overspeed, current and output during recovery.
- [ ] 4. Repeat disturbance steps at approved speeds, directions, supply values and load/inertia corners.
- [ ] 5. In HIL first, request an unreachable but bounded operating point or apply bounded load to saturate control. Hold only for the approved duration.
- [ ] 6. Restore a reachable operating point. Verify stored integral action does not cause excessive overshoot, prolonged saturation or oscillation.
- [ ] 7. Repeat saturation followed by STOP. Measure safe-output time and mechanical stop time/travel separately.
- [ ] 8. Attach recovery traces and metrics; identify whether saturation came from firmware, supply, current regulation or the mechanical load.

PASS CRITERIA

Recovery meets C; output/current remain bounded by E; saturation does not prevent the required stop response or cause a post-release speed excursion beyond C/M.

Result Not Run

Run / method / conditions

Evidence / measurements / deviations / N/A justification

Operator / date

Reviewer / date

HIL-006 | Direction reversal response

Method: H + P | Repeat across the approved Q matrix; use the signed-off E/M/C/T/F/S/B/U limits.

SETUP / WHAT TO WATCH

Capture speed/direction, current, motor/bus voltage, PWM and command timestamps. Use a supply/fixture that can handle the approved regenerated energy.

ACTIONS TO COMPLETE

- [ ] 1. Approve reversal ramp/braking policy, initial speed/load cases, current/regen limits, settling limits and abort thresholds.
- [ ] 2. Start at the lowest approved forward speed and reverse according to the product command sequence.
- [ ] 3. Measure deceleration, zero crossing, reverse acceleration, peak current, bus-voltage rise and final settling.
- [ ] 4. Repeat reverse-to-forward under the same conditions. Confirm direction changes only as commanded.
- [ ] 5. Expand to the approved speed, inertia, load and supply matrix after low-energy cases pass.
- [ ] 6. Issue a replacement or STOP command during reversal. Verify the approved priority/cancellation policy and bounded output.
- [ ] 7. Observe the post-reversal hold period. Verify no stale command or stored control action causes a second unintended reversal.
- [ ] 8. Save traces and compare every reversal metric with approved limits; note fixture energy absorption and any abort.

PASS CRITERIA

Reversal timing, current, regenerated voltage, and speed excursion remain within C/E/M; no uncontrolled reversal or restart.

Result Not Run

Run / method / conditions

Evidence / measurements / deviations / N/A justification

Operator / date

Reviewer / date

HIL-007 | STOP and pending-motion cancellation

Method: H + P | Repeat across the approved Q matrix; use the signed-off E/M/C/T/F/S/B/U limits.

SETUP / WHAT TO WATCH

Observe CAN reception, PWM/driver output, independent shaft speed/position and current on a common timebase.

ACTIONS TO COMPLETE

- [ ] 1. Approve STOP priority, safe electrical state, stop-speed threshold, maximum output latency, stop time/travel and fresh-command policy.
- [ ] 2. Issue STOP while idle. Verify inactive behavior is maintained without unintended movement.
- [ ] 3. Schedule future motion, then STOP before the target tick. Wait beyond that tick and verify the canceled command never actuates.
- [ ] 4. Issue STOP during motion in both directions over the approved speed/load cases. Measure electrical and mechanical stop times separately.
- [ ] 5. Repeat while telemetry is requested and under maximum supported command traffic. Verify STOP meets its deadline.
- [ ] 6. Queue multiple motion commands, issue STOP and continue observing. Confirm stale queued commands do not restart the motor.
- [ ] 7. Repeat STOP during reversal and controller saturation using the approved fixture limits; verify bounded current and travel.
- [ ] 8. After stopping, send a fresh authorized command and verify correct restart. Attach timing, travel and cancellation evidence.

PASS CRITERIA

Safe electrical output reached within T; shaft speed falls below the M stop threshold within the allowed stop time/travel; queued or delayed commands do not restart motion without fresh authorization.

Result Not Run

Run / method / conditions

Evidence / measurements / deviations / N/A justification

Operator / date

Reviewer / date

HIL-008 | CAN addressing and frame validation

Method: H | Repeat across the approved Q matrix; use the signed-off E/M/C/T/F/S/B/U limits.

SETUP / WHAT TO WATCH

Use a timestamped CAN FD generator/analyzer and monitor setpoint, queued motion and output. Record the approved source-ID and payload-target rules.

ACTIONS TO COMPLETE

- [ ] 1. Freeze accepted standard/extended IDs, CAN/CAN-FD flags, BRS policy, DLC and padding rules in the protocol record.
- [ ] 2. Send each supported valid command to the local target. Confirm only the specified operation occurs.
- [ ] 3. Repeat valid broadcast commands. Verify the approved broadcast behavior for motion, STOP and queries.
- [ ] 4. Vary source CAN ID and payload target independently to wrong values. Verify neither setpoint nor scheduled motion changes.
- [ ] 5. Send unsupported frame formats/flags and shorter/longer DLC values at controlled rates. Check the specified rejection behavior.
- [ ] 6. Repeat rejected traffic while moving and with a future move pending. Verify there is no output disturbance or altered schedule.
- [ ] 7. Send a valid command after each invalid group. Verify the interface remains responsive within the specified deadline.
- [ ] 8. Save transmitted/received frames and output traces; list each frame variant and its observed accept/reject result.

PASS CRITERIA

Valid commands produce the specified behavior; rejected traffic changes neither setpoint nor scheduled motion. Rejection handling follows B and never causes unintended actuation.

Result Not Run

Run / method / conditions

Evidence / measurements / deviations / N/A justification

Operator / date

Reviewer / date

HIL-009 | Malformed payload handling

Method: H | Repeat across the approved Q matrix; use the signed-off E/M/C/T/F/S/B/U limits.

SETUP / WHAT TO WATCH

Use a reproducible CAN payload corpus with case identifiers; capture bus traffic, reset causes, output and control responsiveness.

ACTIONS TO COMPLETE

- [ ] 1. Approve handling of unknown actions, reserved fields, invalid numeric values and boundary cases. Define the safe test envelope.
- [ ] 2. Send unknown action/subaction values and inconsistent command fields. Verify rejection or explicitly specified bounded handling.
- [ ] 3. Exercise speed values at, below and above valid bounds. Include NaN, infinities and invalid sign/direction combinations.
- [ ] 4. Exercise truncated logical payloads, byte-order errors and reserved/padding patterns using valid bus framing where required.
- [ ] 5. Repeat malformed cases during idle, motion and pending schedules. Confirm they cannot introduce or change motion unexpectedly.
- [ ] 6. While valid motion commands are withheld, continue malformed traffic. Verify it does not refresh motion authorization.
- [ ] 7. Run the approved seeded malformed-traffic campaign. Monitor resets, lockups, memory diagnostics when available and physical output bounds.
- [ ] 8. Restore valid traffic and verify normal responsiveness. Save the corpus, seed, frame log and any failing case for replay.

PASS CRITERIA

Rejection or bounded handling matches the approved protocol; no uncontrolled output, memory corruption, reset, or lockup; malformed commands do not refresh motion authorization.

Result Not Run

Run / method / conditions

Evidence / measurements / deviations / N/A justification

Operator / date

Reviewer / date

HIL-010 | CAN burst and queue overload behavior

Method: H | Repeat across the approved Q matrix; use the signed-off E/M/C/T/F/S/B/U limits.

SETUP / WHAT TO WATCH

Capture ingress/egress CAN timing, accepted commands, motion output and queue diagnostics where available. Use deterministic traffic profiles.

ACTIONS TO COMPLETE

- [ ] 1. Approve sustained rate, burst length, bus utilization, overflow/replacement policy and response/STOP deadlines.
- [ ] 2. Apply sustained valid traffic up to the rated limit while controlling the motor. Confirm accepted operations and responses remain correct.
- [ ] 3. Apply maximum allowed bursts with mixed MOVE, STOP, broadcast and query traffic. Verify ordering/replacement and latency.
- [ ] 4. Increase traffic beyond the supported envelope in controlled steps. Verify the declared overflow response and bounded output.
- [ ] 5. Send STOP during saturation of traffic/resources. Measure its response or verify the specified independent command-loss protection.
- [ ] 6. Stop valid motion updates while other traffic continues. Confirm stale commands cannot keep the motor authorized.
- [ ] 7. Return to supported traffic. Verify recovery without reset, stuck queues, replay of stale motion or lost readiness.
- [ ] 8. Record sent/accepted/rejected counts, worst response times, bus load, output traces and the exact replay profile.

PASS CRITERIA

Within B: no lost accepted commands and response/STOP deadlines met. Beyond B: declared overflow policy activates, output stays bounded, and STOP or command-loss protection remains effective.

Result Not Run

Run / method / conditions

Evidence / measurements / deviations / N/A justification

Operator / date

Reviewer / date

HIL-011 | Command payload integrity and ordering

Method: H | Repeat across the approved Q matrix; use the signed-off E/M/C/T/F/S/B/U limits.

SETUP / WHAT TO WATCH

Record uniquely distinguishable command payloads, timestamps, target ticks and resulting motion. Use independent expected-sequence tracking.

ACTIONS TO COMPLETE

- [ ] 1. Approve FIFO/replacement, duplicate-command and cancellation behavior; define when a command counts as accepted.
- [ ] 2. Send alternating commands with clearly different valid speed, direction and target-tick values while previous work is pending.
- [ ] 3. Burst the sequence within rated limits. Verify each executed action uses one complete intended payload, never a mixture.
- [ ] 4. Update a scheduled move before it fires. Verify the approved replace/retain policy and correct target data.
- [ ] 5. Repeat duplicate commands and equal-target-tick cases. Check exactly the specified number and order of operations.
- [ ] 6. Insert STOP and error/recovery events into the sequence. Confirm canceled commands cannot execute afterward.
- [ ] 7. Repeat near queue capacity and while telemetry is active. Verify behavior remains deterministic within the rated envelope.
- [ ] 8. Save an expected-versus-observed operation table with raw bus/output traces; include seeds and sequence identifiers.

PASS CRITERIA

Observed action ordering and values match the approved queue/replacement policy; no payload mixing, obsolete execution after STOP, or duplicate actuation.

Result Not Run

Run / method / conditions

Evidence / measurements / deviations / N/A justification

Operator / date

Reviewer / date

HIL-012 | Scheduled motion and counter timing

Method: H | Repeat across the approved Q matrix; use the signed-off E/M/C/T/F/S/B/U limits.

SETUP / WHAT TO WATCH

Use a shared timebase for CAN reception, target-counter interpretation and first physical drive change. Measure clock offset and uncertainty.

ACTIONS TO COMPLETE

- [ ] 1. Approve tick units, timebase synchronization, minimum lead time, scheduling horizon, wrap rules and late/duplicate-command policy.
- [ ] 2. Schedule several valid future starts. Measure first drive action against each target and check early/late error.
- [ ] 3. Test just-inside and just-outside the minimum lead time, then past targets. Verify the specified reject/execute behavior.
- [ ] 4. Replace a pending schedule and repeat duplicate targets. Confirm correct cancellation or retention according to the protocol.
- [ ] 5. Exercise counter wraparound with targets on both sides of rollover. Verify no unexpectedly early, late or repeated start.
- [ ] 6. Reset the counter with an alarm pending. Verify its required cancellation/rebasing behavior, then test a fresh schedule.
- [ ] 7. Repeat with maximum supported traffic and representative control load. Confirm scheduling error stays within the approved tolerance.
- [ ] 8. Save target/actual tick pairs, capture uncertainty, load conditions and cancellation traces for every timing case.

PASS CRITERIA

Valid starts satisfy T scheduling tolerance with no early motion beyond that tolerance. Rejected or canceled schedules never actuate. Counter reset invalidates/rebases pending work exactly as specified.

Result Not Run

Run / method / conditions

Evidence / measurements / deviations / N/A justification

Operator / date

Reviewer / date

HIL-013 | Telemetry during operation

Method: H + P | Repeat across the approved Q matrix; use the signed-off E/M/C/T/F/S/B/U limits.

SETUP / WHAT TO WATCH

Log timestamped requests/responses, an independent counter/speed reference, control timing and motor output.

ACTIONS TO COMPLETE

- [ ] 1. Approve response types, byte order, scaling, freshness, request correlation and response deadlines before testing.
- [ ] 2. Request each supported response while idle. Compare every field against a known reference and check frame formatting.
- [ ] 3. Repeat during steady motion in both directions. Verify queries do not alter setpoint, motion authorization or required output behavior.
- [ ] 4. Issue requests during scheduled starts, STOP and normal state changes. Check defined state-dependent response behavior.
- [ ] 5. Increase query rate to the supported limit while maintaining motion commands. Check latency, data freshness and control deadlines.
- [ ] 6. Create bounded TX congestion or temporary response failure. Verify no indefinite control blockage or unintended actuation.
- [ ] 7. Restore normal bus conditions. Confirm the approved recovery behavior and no stale response mistaken for current data.
- [ ] 8. Attach request/response comparisons, worst latency/freshness and concurrent speed/PWM traces. Mark unimplemented responses Blocked.

PASS CRITERIA

Response contents, byte order, freshness, and deadline match B/protocol; no unintended stop, setpoint change, or missed control deadline. Unimplemented response types remain work in progress.

Result Not Run

Run / method / conditions

Evidence / measurements / deviations / N/A justification

Operator / date

Reviewer / date

HIL-014 | State/event transition coverage

Method: H | Repeat across the approved Q matrix; use the signed-off E/M/C/T/F/S/B/U limits.

SETUP / WHAT TO WATCH

Use a controlled event stimulus and observe state plus actual output, queued work and responses. Internal hooks require documented equivalence to real events.

ACTIONS TO COMPLETE

- [ ] 1. Approve a state/event matrix for all five states, including ignored, rejected and error events. Do not use existing WIP code as the acceptance oracle.
- [ ] 2. Enter each reachable state using a defined precondition sequence; record how transient states are exercised.
- [ ] 3. Apply every supported event in each state, including repeated and unexpected events. Record the expected and observed next state/action.
- [ ] 4. Exercise events arriving during entry/exit work, scheduled motion and query handling. Verify queue and output side effects.
- [ ] 5. Check ignored/rejected cases preserve the required motion and output state; verify telemetry behavior matches the approved specification.
- [ ] 6. Check STOP, reset and error priorities when events coincide or arrive in quick succession, using repeatable ordering.
- [ ] 7. Identify unreachable/unimplemented paths as Blocked or justified N/A rather than passing them by inference.
- [ ] 8. Attach the completed coverage matrix, missing-case disposition and synchronized output/event evidence; all applicable pairs must be accounted for.

PASS CRITERIA

State, physical output, scheduled work, and response match the specification. WIP paths in the design remain pending, not accepted behavior. A telemetry query must preserve commanded motion unless the product specification explicitly requires otherwise.

Result Not Run

Run / method / conditions

Evidence / measurements / deviations / N/A justification

Operator / date

Reviewer / date

HIL-015 | Real-time deadlines under load

Method: H + P | Repeat across the approved Q matrix; use the signed-off E/M/C/T/F/S/B/U limits.

SETUP / WHAT TO WATCH

Capture loop execution markers, CAN reception, PWM changes, alarm events and output using a common timebase; record instrumentation overhead.

ACTIONS TO COMPLETE

- [ ] 1. Approve measured loop-period/jitter limits, command/STOP deadlines, allowed miss count and observation duration.
- [ ] 2. Measure an idle-bus control baseline at representative speeds/loads with production logging settings.
- [ ] 3. Increase CAN RX/TX traffic to rated sustained load and bursts while continuing motor control and timer alarms.
- [ ] 4. Combine worst supported telemetry, logging and event processing. Measure period distributions and maximum latency, not only averages.
- [ ] 5. Inject scheduled starts and STOP into the loaded runs. Verify each critical deadline under the same workload.
- [ ] 6. Repeat over approved supply/load and thermal cases. Check output/current bounds as well as timing.
- [ ] 7. Repeat representative measurements with intrusive debug hooks removed or quantify their effect; verify HIL has no timing overruns.
- [ ] 8. Save timestamp traces, observation duration, distributions, worst cases and deadline-miss counts against T/U limits.

PASS CRITERIA

Measured loop period/jitter, command latency, and deadline-miss count satisfy T throughout Q duration. Report maximum and distribution, not only averages.

Result Not Run | Run / method / conditions

Evidence / measurements / deviations / N/A justification

Operator / date | Reviewer / date

HIL-016 | Quadrature decoding and velocity estimation

Method: H | Repeat across the approved Q matrix; use the signed-off E/M/C/T/F/S/B/U limits.

SETUP / WHAT TO WATCH

Connect a calibrated A/B generator to PA6/PA7 with compatible levels. Observe injected edges and decoded count/speed on aligned timestamps.

ACTIONS TO COMPLETE

- [ ] 1. Approve encoder scaling, count direction, maximum edge rate, velocity window and quantization/error limits.
- [ ] 2. Inject a known forward count sequence, then reverse. Verify exact signed count change and expected angular displacement.
- [ ] 3. Inject one output revolution and multiple revolutions; compare against the approved 2448-count baseline or revised scaling.
- [ ] 4. Sweep from low-speed edge spacing to maximum supported rate. Check count loss, sign and velocity estimation error.
- [ ] 5. Reverse near zero and traverse count rollover. Verify continuous, correctly signed interpretation according to the specification.
- [ ] 6. Inject valid edge jitter/quantization patterns and standstill intervals. Check allowed velocity ripple without confusing valid input with faults.
- [ ] 7. Compare firmware velocity with the generator's known trajectory over the approved measurement window, both directions.
- [ ] 8. Save edge/count reference data, speed-error/ripple results and capture uncertainty for each rate and rollover case.

PASS CRITERIA

Decoded count equals injected count within the declared capture uncertainty; angular scaling agrees with 2,448 counts/revolution or the approved revision. Velocity estimation meets C over the defined sampling window.

Result Not Run

Run / method / conditions

Evidence / measurements / deviations / N/A justification

Operator / date

Reviewer / date

HIL-017 | Encoder and feedback faults

Method: H, then representative P | Repeat across the approved Q matrix; use the signed-off E/M/C/T/F/S/B/U limits.

SETUP / WHAT TO WATCH

Use controlled A/B fault insertion with independent knowledge of true plant/shaft motion. Start in HIL before representative low-energy powered cases.

ACTIONS TO COMPLETE

- [ ] 1. Approve which faults are detectable, detection thresholds/timeouts and safe responses, including behavior at genuine standstill.
- [ ] 2. During motion, freeze both feedback channels. Measure detection time and time to the specified safe output.
- [ ] 3. Open each channel separately, then force supported stuck-high/stuck-low cases through the rated fixture. Verify the expected response.
- [ ] 4. Inject intermittent pulses, implausibly high edge rates and invalid phase patterns. Check bounded motor output and detection behavior.
- [ ] 5. Reverse feedback phase/sign relative to true motion in HIL. Verify the controller cannot sustain uncontrolled acceleration.
- [ ] 6. Repeat valid standstill and low-speed conditions. Confirm no unacceptable false fault under the approved policy.
- [ ] 7. Remove each injected fault and follow the recovery sequence. Verify no automatic replay of stale motion.
- [ ] 8. Record fault-onset, detection, safe-output and shaft-stop timestamps with evidence of true motion; document unavailable diagnostics as Blocked.

PASS CRITERIA

Each detectable fault is identified within F and produces S within its deadline; behavior at true standstill is distinguished from loss of feedback. No sustained acceleration toward an invalid feedback target.

Result Not Run

Run / method / conditions

Evidence / measurements / deviations / N/A justification

Operator / date

Reviewer / date

HIL-018 | Loss of motion-command freshness

Method: H + P | Repeat across the approved Q matrix; use the signed-off E/M/C/T/F/S/B/U limits.

SETUP / WHAT TO WATCH

Record valid motion commands, all background CAN traffic, PWM/current and shaft speed. Identify exactly which command refreshes authorization.

ACTIONS TO COMPLETE

- [ ] 1. Approve freshness timeout, last-valid-command reference point, stop response and restart authorization policy.
- [ ] 2. Establish controlled motion, then stop sending qualifying motion commands. Measure timeout and resulting electrical/mechanical stop.
- [ ] 3. Repeat with unrelated valid telemetry traffic continuing. Verify it cannot keep stale motion authorized.
- [ ] 4. Repeat with malformed and wrong-address frames continuing. Verify these frames cannot refresh the timeout.
- [ ] 5. Physically interrupt the CAN connection using the fixture and repeat the loss case. Check the same required stop bounds.
- [ ] 6. Test command arrivals just before and after the freshness boundary. Verify documented boundary behavior without indefinite extension.
- [ ] 7. Restore the bus without sending a fresh authorization; then send the approved fresh command sequence. Verify restart only at the allowed point.
- [ ] 8. Save last-valid-command, fault/timeout, output and shaft-stop times; repeat at approved speed/load and traffic corners.

PASS CRITERIA

Timeout measured from the last qualifying command satisfies F; output and shaft stop satisfy S/T/M. Other traffic cannot keep stale motion authorized.

Result Not Run

Run / method / conditions

Evidence / measurements / deviations / N/A justification

Operator / date

Reviewer / date

HIL-019 | CAN faults and recovery

Method: H | Repeat across the approved Q matrix; use the signed-off E/M/C/T/F/S/B/U limits.

SETUP / WHAT TO WATCH

Use a controlled CAN fault generator/fixture capable of missing-ACK and bus-off cases. Capture bus state, retries, control timing and motor output.

ACTIONS TO COMPLETE

- [ ] 1. Approve fault detection/recovery limits and safe-state behavior; distinguish network outage from ordinary frame rejection.
- [ ] 2. Suppress acknowledgments during permitted transmissions. Verify bounded waits/retries and no control-task blockage.
- [ ] 3. Apply controlled bus errors until the intended bus-off condition is reached. Confirm the observed state independently of firmware logs.
- [ ] 4. Exercise RX interruption and TX resource failures with motion or a scheduled move active. Verify the specified protection response.
- [ ] 5. Check STOP or command-loss protection remains effective during the fault, using the specified available path.
- [ ] 6. Restore correct bus conditions. Measure recovery time and verify stale queued movement is not replayed.
- [ ] 7. Send fresh authorized commands after recovery and verify normal operation without hidden pending work.
- [ ] 8. Attach fault settings, bus/error traces, worst control latency, recovery timing and physical-output evidence.

PASS CRITERIA

Fault response meets F/S; control is not blocked by TX waits. Recovery meets F and requires the approved fresh-authorization policy; stale queued motion is not replayed.

Result Not Run

Run / method / conditions

Evidence / measurements / deviations / N/A justification

Operator / date

Reviewer / date

HIL-020 | Controller execution failure and watchdog

Method: H, then low-energy P | Repeat across the approved Q matrix; use the signed-off E/M/C/T/F/S/B/U limits.

SETUP / WHAT TO WATCH

Provide independent capture and an energy-removal control. Document fault hooks, watchdog clock/source and behavior while the debugger is connected.

ACTIONS TO COMPLETE

- [ ] 1. Approve the response deadline and safe output for each task stall, event-processing stall, MCU hang and reset case.
- [ ] 2. In HIL, stall the motor-control task through the approved hook. Verify independent protection acts even though normal control cannot progress.
- [ ] 3. Block event processing separately. Verify protection does not depend solely on the blocked queue/task.
- [ ] 4. Exercise an MCU hang or execution lockup using an approved method. Confirm watchdog/output inhibit remains active and reaches S within F.
- [ ] 5. Repeat an MCU reset during motion. Capture pin states throughout reset and startup, not only after logging resumes.
- [ ] 6. Verify stale commands and pending alarms cannot restart motion following recovery; require the approved fresh authorization.
- [ ] 7. Repeat representative cases on a guarded low-energy motor fixture after HIL passes. Keep measurement/control independent of failed firmware.
- [ ] 8. Save fault trigger, watchdog/reset cause, output and shaft-stop timing; classify any missing independent protection as Blocked/Fail as appropriate.

PASS CRITERIA

An independent watchdog/output-inhibit mechanism achieves the fault-specific S state within F even when normal task execution cannot continue. Restart follows S. A watchdog dependent on the failed task path alone is insufficient evidence.

Result Not Run

Run / method / conditions

Evidence / measurements / deviations / N/A justification

Operator / date

Reviewer / date

HIL-021 | Stall, overload and power-stage protection

Method: P; H for firmware response logic | Repeat across the approved Q matrix; use the signed-off E/M/C/T/F/S/B/U limits.

SETUP / WHAT TO WATCH

Use a rated guarded load/fault fixture with current, voltage and thermal capture. Record available firmware sensing separately from hardware protection.

ACTIONS TO COMPLETE

- [ ] 1. Approve continuous/peak current and duration, overload/stall exposures, thermal/energy cutoffs and recovery policy; verify independent abort operation.
- [ ] 2. Characterize normal startup and loaded current first. Confirm the supply limit does not mask the protection mechanism being evaluated.
- [ ] 3. Apply controlled increasing load through the approved overload points. Observe current regulation, speed and thermal trend.
- [ ] 4. Perform a controlled mechanical stall only within the approved exposure and fixture limits. Measure peak/RMS current and protection timing.
- [ ] 5. If short-circuit testing is required, use only the approved rated fault-insertion procedure; otherwise record this subcase as blocked or out of scope with justification.
- [ ] 6. Remove the fault and observe hardware auto-recovery and system outputs. Verify no unauthorized return to energized motion.
- [ ] 7. Test firmware detection/reporting through actual sensing or a documented HIL equivalent. Do not infer firmware detection from IC shutdown alone.
- [ ] 8. Attach current/voltage/temperature traces, exposure durations and post-test functional checks; record any abort and inspect before reuse.

PASS CRITERIA

Current and temperature stay within approved E limits for the defined exposure, protection acts as specified, and post-fault behavior follows S. Distinguish current regulation from overcurrent shutdown. Record hardware-only protection separately from firmware fault detection.

Result Not Run

Run / method / conditions

Evidence / measurements / deviations / N/A justification

Operator / date

Reviewer / date

HIL-022 | Supply sweep, brownout and restoration

Method: P | Repeat across the approved Q matrix; use the signed-off E/M/C/T/F/S/B/U limits.

SETUP / WHAT TO WATCH

Use a programmable supply/fixture and capture MCU rail, motor rail, reset, driver inputs, current and shaft motion; handle regenerative energy.

ACTIONS TO COMPLETE

- [ ] 1. Approve nominal/min/max operating voltages, disturbance profiles, out-of-range test envelope and stop/restart policy.
- [ ] 2. Sweep the declared operating range at representative load/speed. Verify control and electrical performance within approved limits.
- [ ] 3. Apply defined supply dips with specified depth, duration and slew rate. Measure output behavior and any reset/protective response.
- [ ] 4. Interrupt and restore motor supply during motion, then repeat with a scheduled move pending. Verify no stale motion on restoration.
- [ ] 5. Exercise approved MCU-rail brownout and power-sequencing cases. Capture outputs while firmware cannot run.
- [ ] 6. Repeat near specified thresholds with bounded repetitions. Check for unwanted output cycling or repeated unintended starts.
- [ ] 7. Restore nominal power, wait without fresh authorization, then apply the approved restart sequence.
- [ ] 8. Save rail/output traces, disturbance settings, reset causes and stop/restart measurements; do not exceed the approved fault-test envelope.

PASS CRITERIA

Within E: required performance maintained. Outside the operating range but within the approved fault-test envelope: outputs follow S, no unsafe transient/restart, and fresh authorization is required after recovery.

Result Not Run | Run / method / conditions

Evidence / measurements / deviations / N/A justification

Operator / date | Reviewer / date

HIL-023 | Thermal envelope and overtemperature response

Method: P; H for available sensor logic | Repeat across the approved Q matrix; use the signed-off E/M/C/T/F/S/B/U limits.

SETUP / WHAT TO WATCH

Use controlled ambient/load conditions and calibrated temperature sensing at declared locations. Record the relation and uncertainty between measured surface and junction temperature.

ACTIONS TO COMPLETE

- [ ] 1. Approve ambient range, load duration, thermal equilibrium criterion, temperature limits and abort/cooldown procedure.
- [ ] 2. Run representative continuous loads at nominal ambient until the defined equilibrium or dwell criterion is reached.
- [ ] 3. Repeat approved ambient/load/supply corners. Record speed regulation, current and thermal trend throughout.
- [ ] 4. Check operation stays within declared continuous limits without relying on protective shutdown as normal regulation.
- [ ] 5. For the approved overtemperature case, use controlled thermal stimulus within fixture limits and observe output protection and system response.
- [ ] 6. Verify any firmware temperature/fault diagnostic uses a real sensing path; separately exercise its logic in HIL when applicable.
- [ ] 7. Cool the assembly and observe recovery. Verify repeated thermal recovery cannot cause unauthorized energized motion.
- [ ] 8. Attach temperature-location/calibration records, dwell data, electrical traces and post-test functional results; retain aborted runs.

PASS CRITERIA

Continuous operation meets E/C without thermal-limit violation; protective action and recovery meet F/S. No uncontrolled cycling of energized motion during driver automatic recovery. Sensor-dependent diagnostics require actual sensing evidence.

Result Not Run

Run / method / conditions

Evidence / measurements / deviations / N/A justification

Operator / date

Reviewer / date

HIL-024 | Fault indication and restart authorization

Method: H + P | Repeat across the approved Q matrix; use the signed-off E/M/C/T/F/S/B/U limits.

SETUP / WHAT TO WATCH

Observe physical output, CAN status, RGB/local indication, event state and reset causes with a shared timeline.

ACTIONS TO COMPLETE

- [ ] 1. Approve the fault/clear/rearm state sequence, indication deadlines, reporting on reconnect and fresh-command definition.
- [ ] 2. Trigger each representative supported fault. Verify local/CAN indications agree with the physical-output state within the specified time.
- [ ] 3. Clear only the physical fault without reauthorization. Confirm that motion remains inhibited where the policy requires it.
- [ ] 4. Try stale motion commands, counter reset, communication restoration and controller reset independently. Verify none bypasses the recovery policy.
- [ ] 5. Vary the order of fault clearing, rearming and new motion commands. Verify only an approved sequence enables motion.
- [ ] 6. Repeat with CAN unavailable during the fault. Confirm local protection still works and reconnect reporting follows the protocol.
- [ ] 7. Execute a valid recovery and a fresh motion command followed by STOP. Confirm correct operation and cleared/persistent diagnostic state.
- [ ] 8. Save a sequence/result matrix with indication/output timestamps and any rejected or pending commands.

PASS CRITERIA

Cleared faults alone do not restart motion; only the approved recovery sequence does. Status/RGB/CAN indication matches state within B/F. When CAN is unavailable, local protection still operates and fault reporting follows the defined reconnection policy.

Result Not Run

Run / method / conditions

Evidence / measurements / deviations / N/A justification

Operator / date

Reviewer / date

HIL-025 | Endurance, repeated cycles and resource stability

Method: H + P | Repeat across the approved Q matrix; use the signed-off E/M/C/T/F/S/B/U limits.

SETUP / WHAT TO WATCH

Log command cycles, output/speed/current, thermal trend, reset causes and available memory/thread/queue metrics without exceeding production logging load.

ACTIONS TO COMPLETE

- [ ] 1. Approve duration, repetition count, speed/load profile, fault cycles, drift limits and resource/timing limits; freeze the automated sequence.
- [ ] 2. Run sustained operation at representative rated conditions. Record control/timing/thermal trends throughout the full duration.
- [ ] 3. Execute repeated start, STOP, reversal and query cycles at the approved rates and operating corners.
- [ ] 4. Combine supported maximum traffic with the cycle profile. Check correct responses and no gradual queue/resource buildup.
- [ ] 5. Insert controlled fault/recovery cycles according to the plan. Verify each recovery still requires the defined authorization.
- [ ] 6. Periodically repeat a reference speed step and STOP measurement. Compare against initial response and approved drift limits.
- [ ] 7. After the run, perform a normal startup, command, telemetry and STOP check. Retain failures and partial/aborted runs.
- [ ] 8. Archive the replay script/seed, full duration log, cycle counts, extrema, resets and resource trends; do not extrapolate beyond the tested duration.

PASS CRITERIA

No unexpected resets, deadlocks, resource exhaustion, data corruption, or output outside E/M; timing and control remain within T/C. Capture reset causes, thermal trend, and resource metrics where available.

Result Not Run

Run / method / conditions

Evidence / measurements / deviations / N/A justification

Operator / date

Reviewer / date

HIL-026 | Release-matrix regression and closure

Method: H + P | Repeat across the approved Q matrix; use the signed-off E/M/C/T/F/S/B/U limits.

SETUP / WHAT TO WATCH

Use a controlled configuration matrix covering firmware, board, motor, parameters and test/model revisions. Preserve links to all prior raw evidence.

ACTIONS TO COMPLETE

- [ ] 1. Approve the required sample/corner/repetition matrix and the rule for selecting regression cases after each change.
- [ ] 2. Identify every changed component or parameter and assess affected interfaces, timing, control and protection behavior.
- [ ] 3. Run all applicable selected HIL cases on the release candidate; do not transfer results from a different binary without a reviewed justification.
- [ ] 4. Run the required representative powered cases for the approved board/motor/load matrix.
- [ ] 5. Recheck fault-stop and restart behavior whenever control, scheduling, communications or power hardware changes.
- [ ] 6. Account for every requirement as Pass, Fail, Blocked, Not Run or justified N/A. Resolve acceptance-limit TBDs before a Pass decision.
- [ ] 7. Review failures, missing coverage and changed model/fixture assumptions separately from DUT changes; record disposition and reviewer.
- [ ] 8. Archive the release binary/hash, configuration matrix, signed results and raw evidence. Release only when the approved gate is satisfied.

PASS CRITERIA

Every applicable requirement has passing evidence for Q coverage; changes to the test model or acceptance limits are reviewed separately from DUT changes.

Result Not Run

Run / method / conditions

Evidence / measurements / deviations / N/A justification

Operator / date

Reviewer / date

PI/PID response measurement worksheet

Use with HIL-004, 005, 006 and 015. Duplicate for each operating point and repetition.

HIL ID / run / repetition / trace reference

Control mode / Kp / Ki / Kd / filter settings

Supply / ambient / motor-load-inertia condition

Measured loop period / jitter / feedback scaling

Initial/final reference / load step / stimulus timestamp

METRIC	LIMIT / DEFINITION	MEASURED
Rise time (10-90% of step)		
Settling time / band / hold window		
Overshoot (% of signed step size)		
Steady error (absolute / relative)		
Steady speed ripple		
Peak load-disturbance deviation		
Disturbance recovery time		
Peak current / regenerated bus voltage		
STOP output time / shaft stop / travel		

- [ ] Use signed-step normalization for nonzero steps; use absolute speed error near zero. Freeze settling and averaging windows before measuring.
- [ ] Save target, measured and independent speed, PWM/output, current, voltage and synchronized timing. Record integral/saturation data when available.
- [ ] Test PI now as applicable. If derivative support is intended but unfinished, mark D-specific cases Blocked; when enabled, also test noise and output-kick response.

Evidence, closure and references

Record completion and review separately from test execution. This is a draft procedure pack.

- [ ] Account for HIL-001 through HIL-026 and every required subcase, configuration and repetition. Unchecked steps remain incomplete unless disposition is recorded.
- [ ] Confirm each Pass has approved limits, valid instrumentation/model timing, measured results and retrievable raw evidence.
- [ ] Record Fail, Blocked/TBD, Not Run and N/A distinctly. A fixture abort is not a Pass; retain its data and disposition.
- [ ] Verify all relevant fault-stop and restart failures are resolved before the approved release gate. Identify remaining coverage and requirement decisions.
- [ ] Archive firmware binary/hash, board/motor IDs, configuration, fixture/model version, instruments, protocol/limits revisions and operator/reviewer records.

Review / release disposition

Sources and traceability

Project requirement baseline

docs/HIL_TEST_REQUIREMENTS.md - 26 requirement IDs and acceptance parameter sets.

TI DRV8871 datasheet

Input truth table, timing, electrical limits, protection and automatic recovery.

NI: What is hardware-in-the-loop?

Real controller plus a closed-loop real-time plant model.

MathWorks: PID tracking and disturbance rejection

Response assessment and the tracking/disturbance-rejection tradeoff.

The linked sources support the technical basis; procedure details are project-specific engineering proposals. The acceptance limits remain to be approved. This pack does not assert certification or test completion.